Significant and Clinically Relevant Weight Changes at 12 Weeks With Small Molecule GLP-1RA, GSBR-1290

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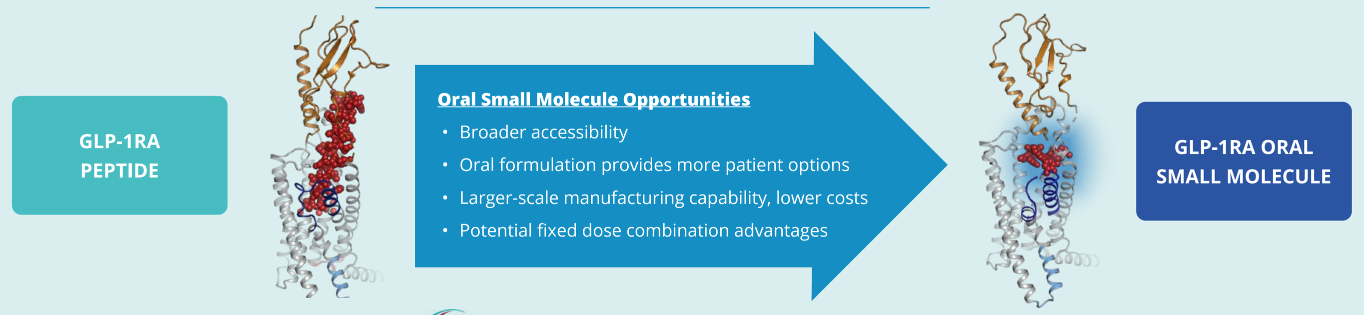
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INTRODUCTION

- Obesity has reached worldwide epidemic proportions, with more than 70% of adults in the United States living with overweight or obesity. Nearly 80% of people with type 2 diabetes mellitus (T2DM) are also living with either overweight or obesity[1]
- Selective glucagon-like peptide receptor agonists (GLP-1RAs) are modeled after naturally occurring GLP-1, which functions to reduce appetite and decrease gastric motility
- Several injectable peptide GLP-1RAs are approved for chronic weight management (CWM) but scalability and accessibility remain a challenge globally
- GSBR-1290 is a novel, oral, non-peptide small molecule GLP-1RA in development for CWM and T2DM, as a potential alternative to injectable peptides

- Using structure-based drug discovery, GSBR-1290 was designed to include specific properties, including ~24-hour drug exposure to enable daily (QD) dosing, low risk of off-target safety effects, and large-scale manufacturing capability

MAKING MEDICINES ACCESSIBLE FOR ALL


GLP-1RA PEPTIDE
Oral Small Molecule Opportunities
Broader accessibility
Oral formulation provides more patient options
Larger-scale manufacturing capability, lower costs
Potential fixed dose combination advantages
GLP-1RA ORAL SMALL MOLECULE

STRUCTURE-BASED DRUG DISCOVERY

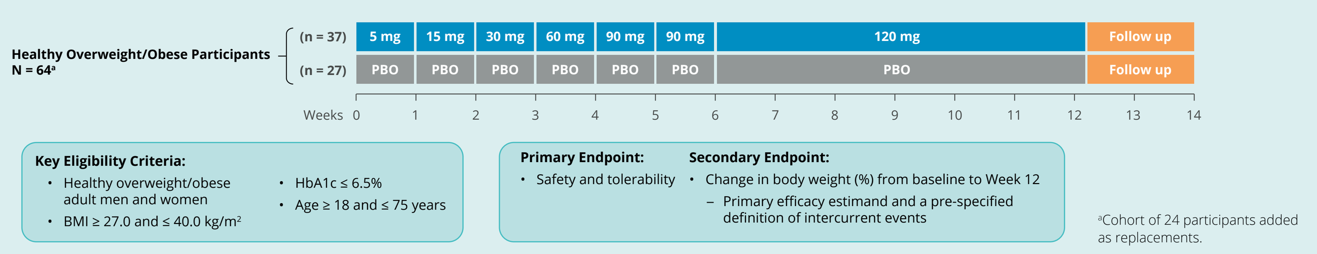
GLP-1RA ORAL SMALL MOLECULE ASPIRATIONAL TARGET PRODUCT PROFILE

- Highly potent, non-peptide, full GLP-1R agonist with minimal β-arrestin signaling engagement
- Weight loss comparable to injectable GLP-1R peptides
- Designed for ~24 hour drug exposure to enable QD dosing in a large patient population
- Clean 6 and 9 month GLP-tox studies with large safety margin (≥ 150 fold at 120 mg)
- Ability to modulate PK properties aimed at optimized tolerability
- Scalability in commercial manufacturing

METHODS

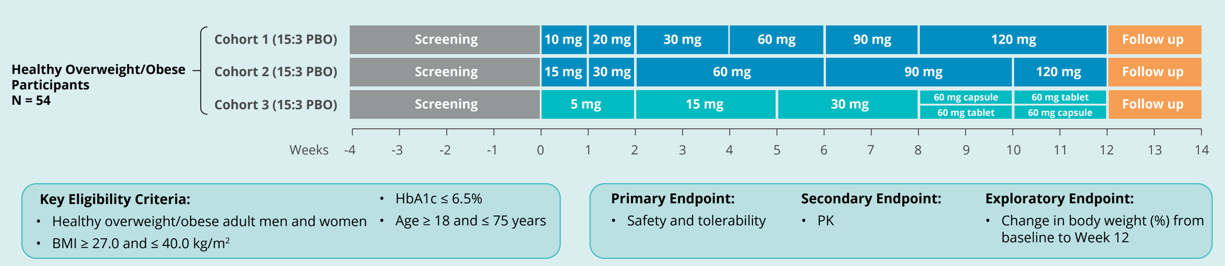
- These studies were designed to evaluate the safety, tolerability, and efficacy of GSBR-1290 in participants living with overweight or obesity
- Both studies included 12 weeks of exposure to GSBR-1290, with 60 or 120 mg/QD as target doses
 - Phase 2a study (GSBR-1290-02)
 - N = 64, randomized 3:1 to GSBR-1290 or placebo
 - Weekly titration scheme, starting at 5 mg up to 120 mg GSBR-1290
 - Capsule to tablet PK study (GSBR-1290-05)
 - N = 54 randomized in 3 cohorts
 - Cohorts with differing titration schemes (1, 2, 3 or 4 week) up to 60 mg or 120 mg to assess the impact on tolerability

FIGURE 1. **GSBR-1290 Phase 2a Study Design**


Healthy Overweight/Obese Participants N = 64[a]
(n = 37) 5 mg 15 mg 30 mg 60 mg 90 mg 90 mg 120 mg Follow up
(n = 27) PBO PBO PBO PBO PBO PBO PBO Follow up
Weeks 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Key Eligibility Criteria:
Healthy overweight/obese adult men and women
BMI ≥ 27.0 and ≤ 40.0 kg/m²
HbA1c ≤ 6.5%
Age ≥ 18 and ≤ 75 years
Primary Endpoint:
Safety and tolerability
Secondary Endpoint:
Change in body weight (%) from baseline to Week 12
Primary efficacy estimand and a pre-specified definition of intercurrent events

[a]Cohort of 24 participants added as replacements.

FIGURE 2. **GSBR-1290 Capsule to Tablet PK Study Design**


Healthy Overweight/Obese Participants N = 54
Cohort 1 (15:3 PBO) Screening 10 mg 20 mg 30 mg 60 mg 90 mg 120 mg Follow up
Cohort 2 (15:3 PBO) Screening 15 mg 30 mg 60 mg 90 mg 120 mg Follow up
Cohort 3 (15:3 PBO) Screening 5 mg 15 mg 30 mg 60 mg capsule 60 mg tablet 60 mg tablet 60 mg capsule Follow up
Weeks -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Key Eligibility Criteria:
Healthy overweight/obese adult men and women
BMI ≥ 27.0 and ≤ 40.0 kg/m²
HbA1c ≤ 6.5%
Age ≥ 18 and ≤ 75 years
Primary Endpoint:
Safety and tolerability
Secondary Endpoint:
PK
Exploratory Endpoint:
Change in body weight (%) from baseline to Week 12

- The combined data from both studies of percent (%) change of body weight from baseline among the obesity cohorts was analyzed using a mixed effects model for repeated measures. The models incorporate treatment group, visit, treatment-by-visit interaction, and baseline values as covariates. The 120 mg group from the phase 2a and two 120 mg groups (cohort 1 and 2) from the capsule to tablet PK study were combined as single 120 mg group. The placebo groups from both studies were combined as one placebo group

RESULTS

TABLE 1. **Baseline Characteristics**

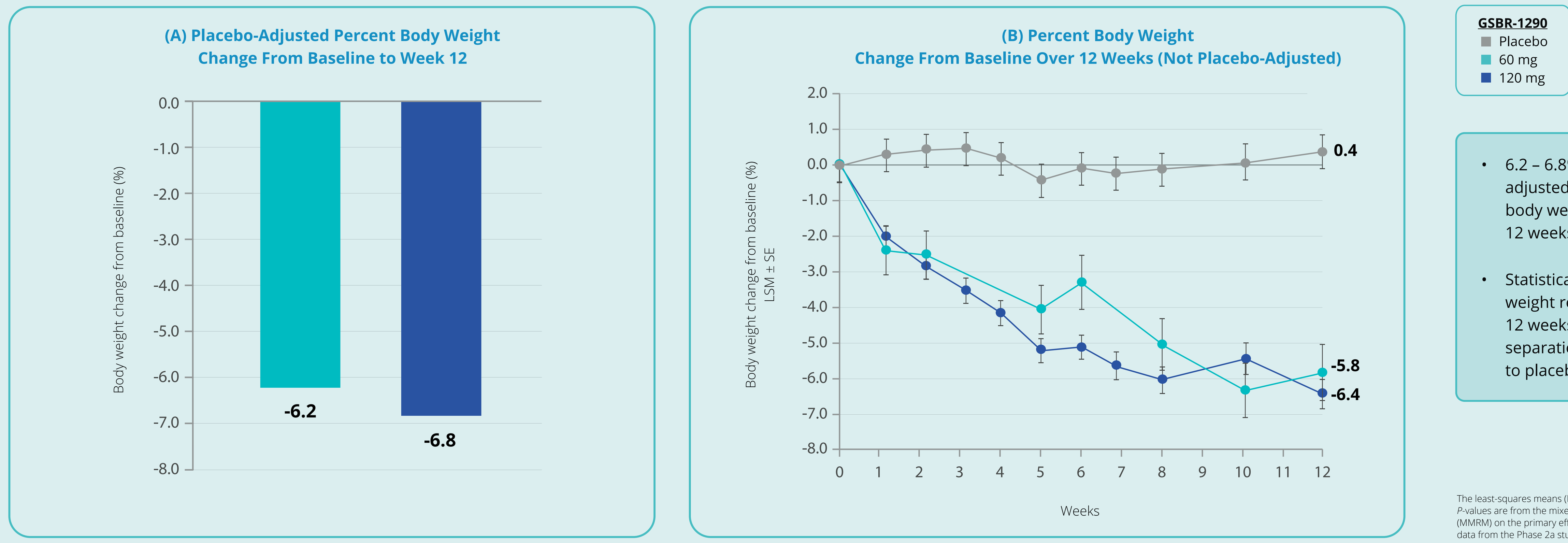
Characteristic Mean (SD) or n (%)	GSBR-1290 60 mg (N = 15)	GSBR-1290 120 mg (N = 67)	Placebo (N = 36)
Age, years	47.6 (11.6)	46.0 (12.8)	45.5 (12.5)
Sex, Female, n (%)	7 (46.7)	36 (53.7)	16 (44.4)
Race, Hispanic or Latino, n (%)	9 (60.0)	36 (53.7)	18 (50.0)
Weight, kg	87.0 (11.6)	88.6 (13.4)	89.1 (13.6)
BMI, kg/m²	29.8 (1.7)	31.0 (3.0)	31.2 (3.0)

TABLE 2. **Participant Disposition**

Number of Participants Reporting N (%)	GSBR-1290 60 mg (N = 15)	GSBR-1290 120 mg (N = 67)	Placebo (N = 36)
Discontinued study not due to Adverse Events (AEs)	2 (13.3)	5 (7.5)	0
Discontinued study due to AEs related to treatment	1 (6.7)	7 (10.4)	2 (5.6)
Dose discontinuation due to AEs	3 (20.0)	7 (10.4)	1 (2.8)

60 and 120mg pooled across studies, includes different titration schemes

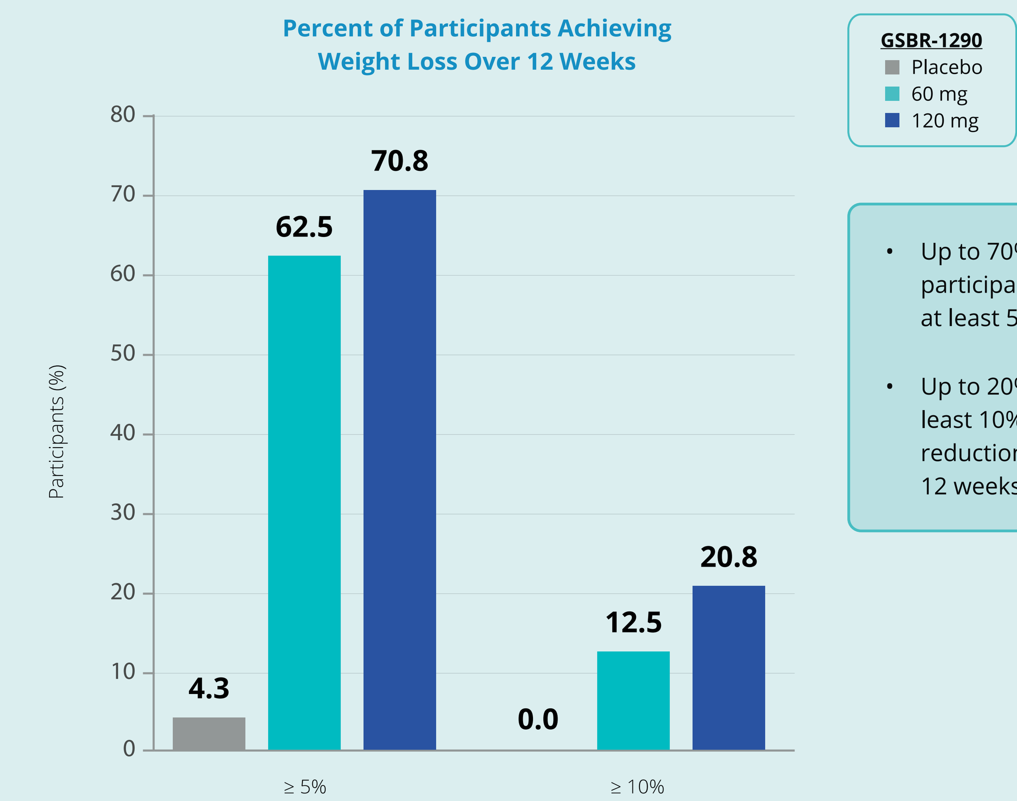
FIGURE 3. **Change in Body Weight Over 12 Weeks (Placebo-Adjusted)**


(A) Placebo-Adjusted Percent Body Weight Change From Baseline to Week 12
Body weight change from baseline (%)
-6.2
-6.8
(B) Percent Body Weight Change From Baseline Over 12 Weeks (Not Placebo-Adjusted)
Body weight change from baseline (%) LSM ± SE
0.4
-5.8
-6.4
Weeks
GSBR-1290
Placebo
60 mg
120 mg

- 6.2 – 6.8% (placebo-adjusted) reduction in body weight across 12 weeks
- Statistically significant weight reduction over 12 weeks with early separation compared to placebo

The least-squares means (LSM), standard errors (SE) and *P*-values are from the mixed model for repeated measures (MMRM) on the primary efficacy estimand for the combined data from the Phase 2a study and capsule to tablet PK study.

FIGURE 4. **Change in Body Weight Over 12 Weeks: Responder Analysis**


Percent of Participants Achieving Weight Loss Over 12 Weeks
Participants (%)
4.3
62.5
70.8
0.0
12.5
20.8
≥ 5%
≥ 10%
GSBR-1290
Placebo
60 mg
120 mg

- Up to 70% of participants achieved at least 5% weight loss
- Up to 20% achieved at least 10% body weight reduction within 12 weeks

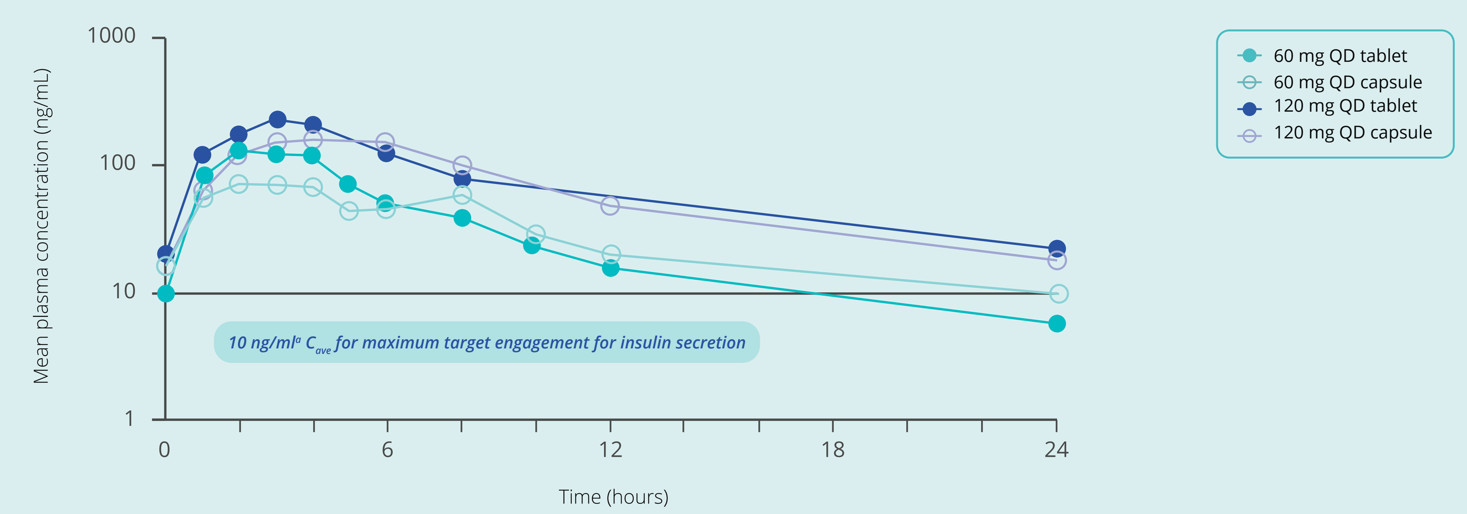
TABLE 3. **Safety: Liver Function Test Profile**

Number of Participants Reporting, N (%)	Phase 2a Study GSBR-1290 120 mg (N = 37)	Phase 2a Study Placebo (N = 27)	Capsule to tablet PK study GSBR-1290 60 mg (N = 15)	Capsule to tablet PK study GSBR-1290 120 mg (N = 30)	Capsule to tablet PK study Placebo (N = 9)
Drug induced liver injury (DILI)	0	0	0	0	0
Hepatic enzymes increased[a]	0	0	0	0	0
Mean Change from Baseline to Day 84	120 mg	Placebo	60 mg	120 mg	Placebo
ALT, (U/L) Mean (SD)	-2.2 (11.7)	-1.4 (8.1)	1.2 (9.3)	0.8 (20.2)	-3.0 (5.1)
AST, (U/L) Mean (SD)	-1.4 (7.2)	-3.0 (11.5)	-0.8 (5.4)	1.7 (31.9)	-1.7 (3.4)
3 or > ULN in ALT or AST	1 (2.7)[b]	1 (3.7)[c]	0	1 (3.3)[d]	0
5 or > ULN in ALT or AST	0	0	0	0	0
10 or > ULN in ALT or AST	0	0	0	0	0

- No DILI or permanent elevations in liver enzymes and no study discontinuations due to elevated liver enzymes

[a]Preferred term in the TEAE reported in the safety database. [b]Female participant with a sporadic increase in ALT (3.9xULN) and AST (2xULN) at day 44, and returned to normal at day 48 without stopping study drug. [c]Male participant with fluctuations in ALT/AST throughout the study with a peak at day 30 (both ALT and AST @3xULN) associated with an increase in creatinine kinase. [d]Male participant with an isolated increase in AST (4XULN) at day 84 associated with an increase in creatinine kinase.

FIGURE 5. **GSBR-1290 Exposure Supports Once Daily (QD) Dosing**


Mean plasma concentration (ng/mL)
10 ng/ml[a] C_{ave} for maximum target engagement for insulin secretion
Time (hours)
60 mg QD tablet
60 mg QD capsule
120 mg QD tablet
120 mg QD capsule

GSBR-1290 Was Designed With Specific Properties Resulting in PK Parameters Supportive of QD Dosing With Broad Safety Margins

Pharmacokinetic Properties		Molecular Properties
• Plasma concentration at 24 hours above 10 ng/mL[a] for 120 mg • Generally proportional exposure between 60 mg and 120 mg	• Large safety margin (≥ 150 fold at 120 mg) with minimal tissue accumulation based on non-clinical data	• Full agonist with minimal ß-arrestin signaling • No glutathione reactivity

[a]10 ng/mL C_{avg} threshold identified through NHP studies that demonstrated maximum insulin secretion

CONCLUSIONS

- Using structure-based drug discovery, GSBR-1290 was designed to be a potent, orally available, safe, non-peptide small molecule GLP-1RA
- Over 12 weeks across both studies, GSBR-1290 showed clinically meaningful and statistically significant body weight reduction in participants with overweight or obesity
 - -6.2 – 6.8% placebo-adjusted weight loss at 60 and 120 mg, respectively, with QD dosing at 12 weeks
 - No new safety findings
 - Low number of study discontinuations due to AEs
- GSBR-1290 PK findings support QD dosing and proportional exposure
- No permanent elevations in liver enzymes. No DILI and no study discontinuations related to liver function
- A 36-week Phase 2b study of GSBR-1290 in participants living with overweight or obesity is expected to initiate in Q4 2024

ABBREVIATIONS

AE = advese event; **CWM** = chronic weight management; **DILI** = drug-induced liver injury; **GLP** = glucagon-like peptide; **GLP-1RA** = glucagon-like peptide receptor agonists; **LSM** = least-squares mean; **MMRM** = mixed model for repeated measures; **NHP** = non human primate; **NOAEL** = no observed adverse effect level; **QD** = once daily; **SE** = standard error; **T2DM** = type 2 diabetes mellitus

REFERENCES

1. Overweight & obesity statistics. NIH, 2021. https://www.niddk.nih.gov/health-information/health-statistics/overweight-obesity. **2.** J Wilding, et al., 2016; *EMA Wegovy Report*, 2021.

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